PROBLEM NS9

Porous Cylinder Quiz with Viscous Heating

Original statement · NS9.pdf

Axisymmetric flow | Suction Reynolds number | Viscous heating | Blowing: no steady solution

OFFICIAL PROBLEM STATEMENT

Consider the steady buoyancy-free motion of a perfect liquid of density ρ, viscosity μ, thermal conductivity k, and specific heat c, induced by an infinitely long porous cylinder of radius R that moves with a constant velocity U_w parallel to its axis, while a constant flow rate $q = 2\pi R V_w$ per unit length is uniformly sucked in the radial direction through the porous wall of the cylinder. Far from the cylinder, $r \to \infty$, the fluid is at rest, at pressure p_∞ and temperature T_∞.

To solve the problem, use a cylindrical coordinate system (r, θ, z), where z is the axis of the cylinder, and assume that the velocity field is axisymmetric, $\partial \bar{v}/\partial \theta = 0$, as well as fully developed, $\partial \bar{v}/\partial z = 0$, thus $\bar{v} = v_r(r)\bar{e}_r + v_z(r)\bar{e}_z$.

1. Using continuity, determine the velocity component $v_r(r)$. (1 point)

2. Obtain the pressure field $p(r, z)$ by integration of the r-momentum equation, demonstrating in particular that $\partial p/\partial z = 0$. (2 points)

3. Using the z-momentum equation, obtain the axial velocity distribution, $v_z(r)$. (2 points)

4. Compute the trajectory of a particle initially placed at $r = R_i > R$ and $z = z_i$, as well as the corresponding sucking time, t_s. (1 point)

5. Calculate the force per unit length that the liquid exerts on the cylinder. (1 point)

6. Write the differential equation that determines the temperature in the liquid $T(r)$. Obtain the function $T(r)$, assuming that the wall is at constant uniform temperature T_w. (2 points)

7. Obtain the conductive heat flux per unit length from the liquid to the wall q_w. (1 point)

8. Show that there is no steady solution if fluid is blown through the wall, instead of being sucked. (1 extra point)

The sheet is headed "INTRODUCTION TO FLUID MECHANICS, QUIZ # 2 (Nov 26, 2012)" and carries no figure.

Source: NS9.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41 and energy Eq. 4.42 in cylindrical coordinates, viscous dissipation ϕ_v and stress components in cylindrical coordinates.

Worked solution

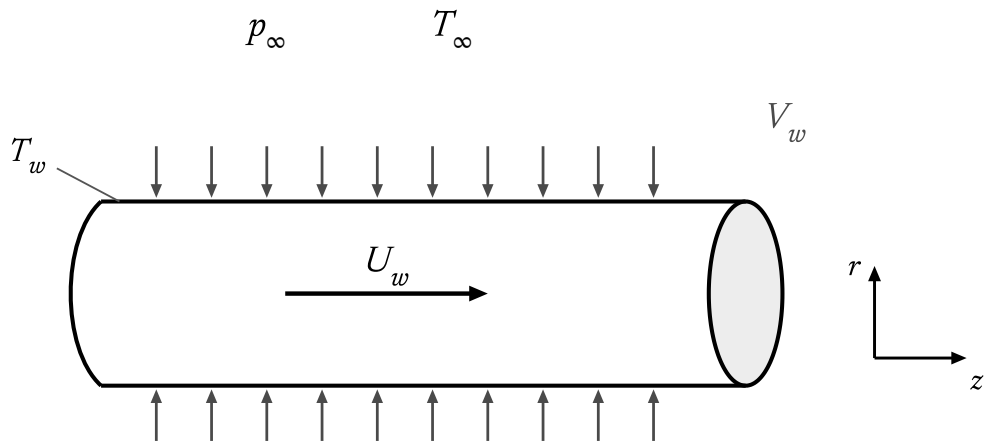
Hypotheses, Data and Unknowns

- Steady, buoyancy-free flow of a perfect liquid (constant ρ, μ, k, c) around an infinitely long cylinder: the fields depend only on r ($\partial/\partial\theta = \partial/\partial z = \partial/\partial t = 0$) and $v_\theta = 0$. No chemical reaction and no radiation.
- The wall is porous: it sucks in the volume flow $q = 2\pi R V_w$ per unit length, uniformly, so $v_r(R) = -V_w$ with $V_w > 0$, and it moves axially with U_w.
- Far from the cylinder the liquid is at rest at p_∞ and T_∞.
- Data: $[\rho,\ \mu,\ k,\ c,\ R,\ U_w,\ V_w,\ p_\infty,\ T_\infty,\ T_w,\ R_i,\ z_i]$. Unknowns: v_r, p, v_z, the trajectory and t_s, the force, $T(r)$ and q_w.
- Dimensionless numbers: the suction Reynolds number $\mathrm{Re} = \rho V_w R/\mu = \rho q/(2\pi\mu)$, the Prandtl number $\mathrm{Pr} = \mu c/k$ and the product $\Pi = \mathrm{Pr}\,\mathrm{Re} = \rho c V_w R/k$.

Symbol	Meaning	SI unit
R	cylinder radius	m
U_w, V_w	axial velocity of the cylinder, suction velocity	m/s
$q = 2\pi R V_w$	suction flow rate per unit length	m^2/s
Re, Pr, Π	suction Reynolds number, Prandtl number, their product	1
T_w, T_∞	wall and ambient temperatures	K
R_i, z_i	initial radial and axial position of the particle	m

Frames, Field Form and Boundary Conditions

- Cylindrical frame (r, θ, z) with z along the axis of the cylinder; the basis $\{\bar{e}_r, \bar{e}_\theta, \bar{e}_z\}$ is orthonormal and right-handed. The liquid occupies $r > R$.
- Field form: $\bar{v} = v_r(r)\,\bar{e}_r + v_z(r)\,\bar{e}_z$, $p = p(r, z)$ until determined, $T = T(r)$.



Setup of NS9 (drawn for the solution): porous cylinder of radius R at temperature T_w moving with U_w along z, suction velocity V_w ($q = 2\pi R V_w$ per unit length), fluid at rest at p_∞, T_∞ far away, axes r, z

Location	Condition	Origin
$r = R$	$v_r = -V_w$	uniform suction through the porous wall
$r = R$	$v_z = U_w$	no-slip on the moving cylinder
$r \to \infty$	$v_z \to 0, p \to p_\infty, T \to T_\infty$	fluid at rest far away
$r = R$	$T = T_w$	uniform wall temperature

Step-by-Step Derivation

3.1 Continuity (item 1)

$$\frac{1}{r}\frac{d(r v_r)}{dr} + \underbrace{\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}}_{v_\theta = 0} + \underbrace{\frac{\partial v_z}{\partial z}}_{\partial_z = 0} = 0 \implies r\, v_r = \text{const} = -R V_w \implies \boxed{v_r(r) = -\frac{R V_w}{r} = -\frac{q}{2\pi r}} \quad (1)$$

3.2 Radial momentum and the pressure (item 2)

The viscous term of the r component vanishes because rv_r is constant, and the θ and z terms are zero:

$$\rho\, v_r \frac{dv_r}{dr} = -\frac{\partial p}{\partial r} \implies p + \frac{\rho v_r^2}{2} = f(z) \tag{2}$$

where integration at fixed z leaves a function of z. The condition $p \to p_\infty$ for $r \to \infty$ (where $v_r \to 0$) holds for every z, so $f(z) = p_\infty$ is a constant:

$$\boxed{p(r) = p_\infty - \frac{\rho R^2 V_w^2}{2r^2}}, \qquad \frac{\partial p}{\partial z} = 0 \tag{3}$$

3.3 Axial momentum (item 3)

The z component of (4.41), with $\partial_z = 0$, $v_\theta = 0$, no mass force and $\partial p/\partial z = 0$, is $\rho v_r\, dv_z/dr = \mu\, r^{-1} d(r\, dv_z/dr)/dr$. Substituting (1) and multiplying by r/μ:

$$\frac{d}{dr}\left(r \frac{dv_z}{dr}\right) + \mathrm{Re}\, \frac{dv_z}{dr} = 0 \tag{4}$$

Try $v_z = r^\alpha$: $\alpha^2 + \mathrm{Re}\,\alpha = 0$, so $\alpha = 0$ or $\alpha = -\mathrm{Re}$ and $v_z = C_1 + C_2 r^{-\mathrm{Re}}$. The condition at infinity gives $C_1 = 0$ and $v_z(R) = U_w$ gives $C_2 = U_w R^{\mathrm{Re}}$:

$$\boxed{v_z(r) = U_w \left(\frac{R}{r}\right)^{\mathrm{Re}}} \tag{5}$$

3.4 Trajectory and sucking time (item 4)

The velocity of the particle is the local velocity. Radial motion: $dr/dt = -RV_w/r$, so $r\,dr = -RV_w\,dt$; integrating from R_i at $t = 0$, $(r^2 - R_i^2)/2 = -RV_w t$:

$$r(t) = \sqrt{R_i^2 - 2RV_w t}, \qquad \boxed{t_s = \frac{R_i^2 - R^2}{2RV_w}} \tag{6}$$

Axial motion: $dz/dr = v_z/v_r = -\dfrac{U_w R^{\mathrm{Re}-1}}{V_w} r^{1-\mathrm{Re}}$. Integrating from R_i (where $z = z_i$), with $\int r^{1-\mathrm{Re}} dr = r^{2-\mathrm{Re}}/(2-\mathrm{Re})$ for $\mathrm{Re} \neq 2$:

$$\boxed{z(r) = z_i + \frac{U_w R}{V_w\,(2-\mathrm{Re})}\left[\left(\frac{R_i}{R}\right)^{2-\mathrm{Re}} - \left(\frac{r}{R}\right)^{2-\mathrm{Re}}\right]} \qquad (\mathrm{Re} \neq 2) \tag{7}$$

For $\mathrm{Re} = 2$, $z = z_i + (U_w R/V_w)\ln(R_i/r)$. The axial position at which the particle is sucked is (7) with $r = R$.

3.5 Force on the cylinder (item 5)

The normal pointing into the liquid is $\bar{n} = +\bar{e}_r$; the pressure and the radial normal stress act radially and integrate to zero around the circumference. The axial shear stress is $\tau'_{rz} = \mu\, dv_z/dr$, and from (5), $dv_z/dr\big|_R = -\mathrm{Re}\,U_w/R$:

$$\tau'_{rz}\Big|_R = -\mu\,\mathrm{Re}\,\frac{U_w}{R} = -\rho V_w U_w, \qquad \boxed{\bar{F} = 2\pi R\,\tau'_{rz}\Big|_R\,\bar{e}_z = -\rho\,q\,U_w\,\bar{e}_z} \tag{8}$$

The force per unit length equals minus the axial momentum given per unit time to the sucked liquid, $\rho q\, U_w$.

3.6 Temperature: the equation (item 6)

The energy equation (4.42) with $v_r = -RV_w/r$, $\partial_z = \partial_\theta = 0$ and the dissipation of the cylindrical expression (Notes.pdf) keeping the non-zero terms, $\phi_v = \mu\left\{2\left[(dv_r/dr)^2 + (v_r/r)^2\right] + (dv_z/dr)^2\right\}$:

$$\rho c\, v_r \frac{dT}{dr} = \phi_v + \frac{k}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right), \qquad \phi_v = \mu\left[\frac{4R^2V_w^2}{r^4} + \frac{\mathrm{Re}^2 U_w^2}{r^2}\left(\frac{R}{r}\right)^{2\mathrm{Re}}\right] \tag{9}$$

since $dv_r/dr = RV_w/r^2$ and $v_r/r = -RV_w/r^2$. Multiplying by r/k and using $\rho c R V_w/k = \Pi$:

$$\frac{d}{dr}\left(r\frac{dT}{dr}\right) + \Pi\,\frac{dT}{dr} = -\frac{\mu}{k}\left[\frac{4R^2V_w^2}{r^3} + \frac{\mathrm{Re}^2 U_w^2 R^{2\mathrm{Re}}}{r^{2\mathrm{Re}+1}}\right] \tag{10}$$

with the boundary conditions $T(R) = T_w$ and $T(\infty) = T_\infty$.

3.7 Temperature: the solution (item 6)

Write $y = dT/dr$, so that (10) is $r\,y' + (1+\Pi)\,y = -S_1 r^{-3} - S_2 r^{-2\mathrm{Re}-1}$ with $S_1 = 4\mu R^2 V_w^2/k$ and $S_2 = \mu\,\mathrm{Re}^2 U_w^2 R^{2\mathrm{Re}}/k$. The homogeneous solution is $y = D\,r^{-1-\Pi}$ (substitute r^s: $s + 1 + \Pi = 0$). For each source term try a power of the same form: $y = A_1 r^{-3}$ gives $r\,y' = -3A_1 r^{-3}$ and $(\Pi - 2)A_1 = -S_1$; $y = A_2 r^{-2\mathrm{Re}-1}$ gives $(\Pi - 2\mathrm{Re})A_2 = -S_2$. Hence, for $\Pi \neq 2$ and $\mathrm{Pr} \neq 2$,

$$A_1 = \frac{S_1}{2-\Pi}, \qquad A_2 = \frac{S_2}{\mathrm{Re}\,(2-\mathrm{Pr})}$$

Integrating y ($\int r^{-n}dr = -r^{1-n}/(n-1)$) and writing $\mu/k = \mathrm{Pr}/c$:

$$T(r) = B - a_1\left(\frac{R}{r}\right)^2 - a_2\left(\frac{R}{r}\right)^{2\mathrm{Re}} + E\left(\frac{R}{r}\right)^{\Pi}, \qquad a_1 = \frac{2\,\mathrm{Pr}\,V_w^2}{c\,(2-\Pi)}, \quad a_2 = \frac{\mathrm{Pr}\,U_w^2}{2c\,(2-\mathrm{Pr})} \tag{11}$$

For $r \to \infty$ every power of R/r vanishes, so $T(\infty) = T_\infty$ gives $B = T_\infty$; $T(R) = T_w$ gives $E = T_w - T_\infty + a_1 + a_2$:

$$\boxed{T(r) = T_\infty - a_1\left(\frac{R}{r}\right)^{2} - a_2\left(\frac{R}{r}\right)^{2\mathrm{Re}} + (T_w - T_\infty + a_1 + a_2)\left(\frac{R}{r}\right)^{\Pi}} \tag{12}$$

The terms a_1 and a_2 are the viscous heating by the radial and by the axial deformation. The cases $\Pi = 2$ and $\mathrm{Pr} = 2$ (resonance of the exponents) give logarithmic terms and are not worked out here.

3.8 Heat flux to the wall (item 7)

The heat flux from the liquid to the wall is $\bar{q}\cdot\bar{n}$ with $\bar{n} = -\bar{e}_r$: $k\,dT/dr\big|_R$. Differentiating (12) and setting $r = R$:

$$\left.\frac{dT}{dr}\right|_R = \frac{1}{R}\left[2a_1 + 2\mathrm{Re}\,a_2 - \Pi\,(T_w - T_\infty + a_1 + a_2)\right] = \frac{1}{R}\left[\Pi\,(T_\infty - T_w) + (2-\Pi)a_1 + (2\mathrm{Re} - \Pi)a_2\right]$$

with $(2-\Pi)a_1 = 2\mathrm{Pr}\,V_w^2/c$ and $(2\mathrm{Re}-\Pi)a_2 = \mathrm{Re}(2-\mathrm{Pr})a_2 = \mathrm{Pr}\,\mathrm{Re}\,U_w^2/(2c)$. Multiplying by k and using $k\,\Pi/R = \rho c V_w$, $k\,\mathrm{Pr}/c = \mu$ and $\mu\,\mathrm{Re}/R = \rho V_w$:

$$k\left.\frac{dT}{dr}\right|_R = \rho c V_w\,(T_\infty - T_w) + \frac{2\mu V_w^2}{R} + \frac{\rho V_w U_w^2}{2}$$

and, multiplying by the area per unit length $2\pi R$,

$$\boxed{q_w = 2\pi R\rho V_w\left[c\,(T_\infty - T_w) + \frac{U_w^2}{2}\right] + 4\pi\mu V_w^2} \tag{13}$$

3.9 Blowing instead of suction (item 8)

If the liquid is blown, $V_w < 0$ and $\mathrm{Re} < 0$. Continuity still gives $v_r = -RV_w/r = R|V_w|/r > 0$, and the equation for v_z is (4) with $\mathrm{Re} < 0$: $v_z = C_1 + C_2\,r^{|\mathrm{Re}|}$. The condition $v_z \to 0$ for $r \to \infty$ requires $C_2 = 0$ (the power grows) and then $C_1 = 0$, so $v_z(R) = U_w$ cannot be satisfied unless $U_w = 0$. Likewise $T = B + D\,r^{-\Pi}$ with $\Pi < 0$ is unbounded unless $D = 0$, which cannot satisfy $T(R) = T_w \neq T_\infty$. Therefore there is no

steady solution with the fluid at rest and at T_∞ far away: the blown liquid carries momentum and heat outward and they cannot be balanced by diffusion towards the cylinder.

Interpretation, Limits and Dimensional Check

- **Physical meaning.** Suction confines the axial momentum to a layer of thickness of order R/Re and the thermal layer to R/Π: the thermal layer is thinner than the momentum layer when $\mathrm{Pr} > 1$. The cylinder pays for the entrained momentum with the force (8), and the wall receives the enthalpy, kinetic-energy and viscous-heating terms of (13).
- **Limits.** For $U_w = 0$, $a_2 = 0$ and E contains only a_1; for negligible dissipation ($a_1 = a_2 = 0$) the solution is $T = T_\infty + (T_w - T_\infty)(R/r)^{\Pi}$ and $q_w = 2\pi R \rho c V_w (T_\infty - T_w)$, the result of NS8. For $V_w \to 0$ the axial profile is uniform and there is no steady solution that vanishes at infinity.
- **Dimensional check.** $\mathrm{Re} = \rho V_w R/\mu$ is dimensionless. $\rho V_w U_w$ has units $\mathrm{N/m^2}$. a_1 and a_2 have units $\mathrm{Pr}\, V_w^2/c = \mathrm{m^2/s^2/(J/(kg\,K))} = \mathrm{K}$. In (13) every term is W/m: $\rho R V_w c\, \Delta T = (\mathrm{kg/m^3})\,\mathrm{m}\cdot\mathrm{m/s}\cdot\mathrm{J/(kg\,K)}\cdot\mathrm{K} = \mathrm{W/m}$ and $\mu V_w^2 = (\mathrm{Pa\,s})(\mathrm{m^2/s^2}) = \mathrm{W/m}$.
- **Energy check.** A steady total-energy balance on the liquid between the wall and infinity gives the same result (13): the energy flux $\rho q\, c T_\infty$ and the pressure work $p_\infty q$ enter at infinity, and the viscous stresses at the wall do the work $4\pi\mu V_w^2 + \rho q\, U_w^2$ on the liquid (from $\tau'_{rr}(R) = 2\mu V_w/R$ and $\tau'_{rz}(R) = -\rho V_w U_w$); through the wall leave $\rho q\,[cT_w + (U_w^2 + V_w^2)/2]$, the pressure work $p(R)\,q$ with $p(R) = p_\infty - \rho V_w^2/2$, and the heat q_w. Hence $\rho q\, cT_\infty + p_\infty q + 4\pi\mu V_w^2 + \rho q U_w^2 = \rho q\,[cT_w + (U_w^2 + V_w^2)/2] + p(R)\,q + q_w$; the radial kinetic energy cancels against $(p_\infty - p(R))\,q = \rho q V_w^2/2$, and what remains is (13).